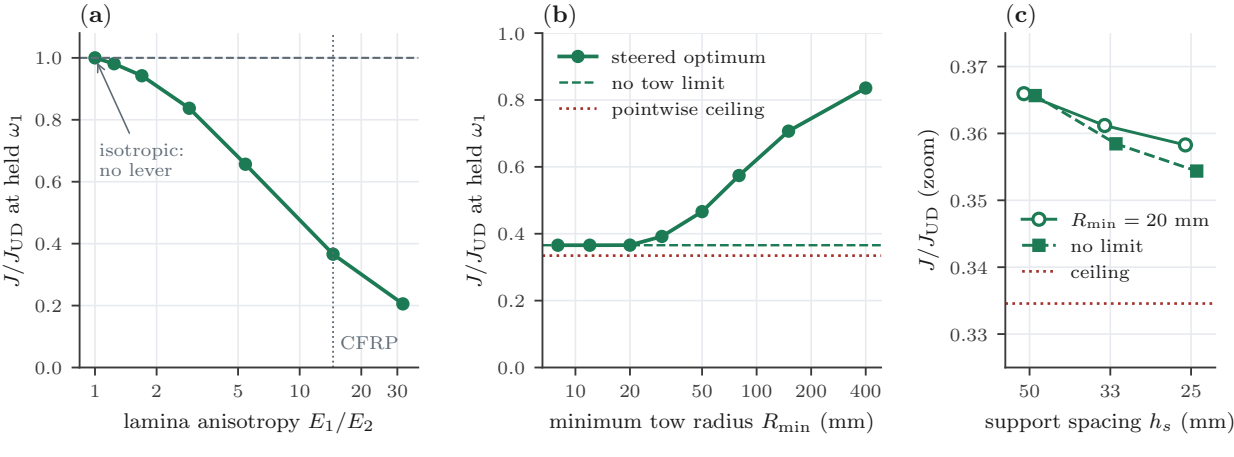

(a)
1.0
0.8
0.6
0.4
0.2
0.0
J/J_{UD} at held ω_1
isotropic:
no lever
CFRP
1
2
5
10
20
30
lamina anisotropy E_1/E_2
(b)
steered optimum
no tow limit
pointwise ceiling
10
20
50
100
200
400
minimum tow radius R_{min} (mm)
(c)
0.37
0.36
0.35
0.34
0.33
J/J_{UD} (zoom)
$R_{\mathrm{min}} = 20$ mm
no limit
ceiling
50
33
25
support spacing h_s (mm)